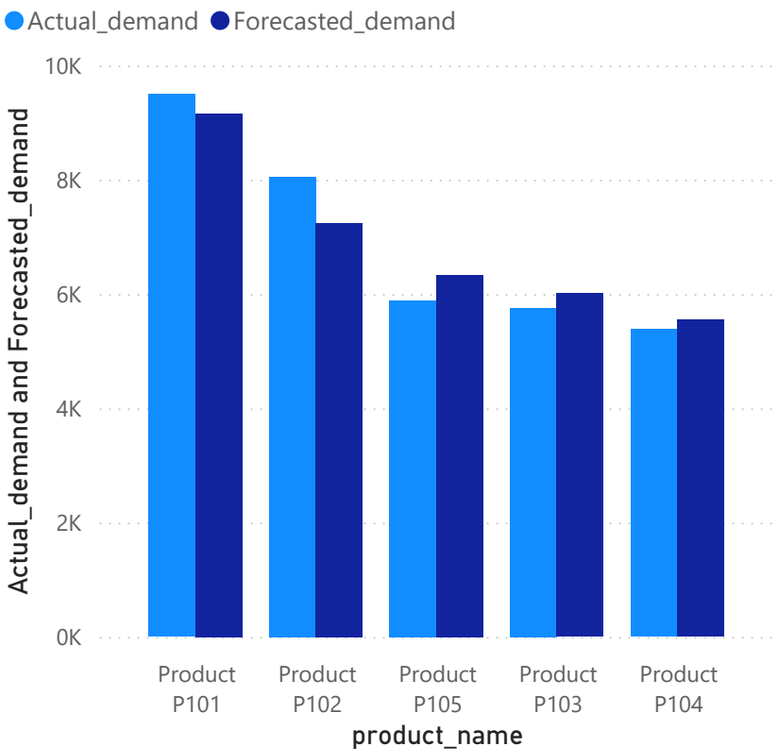
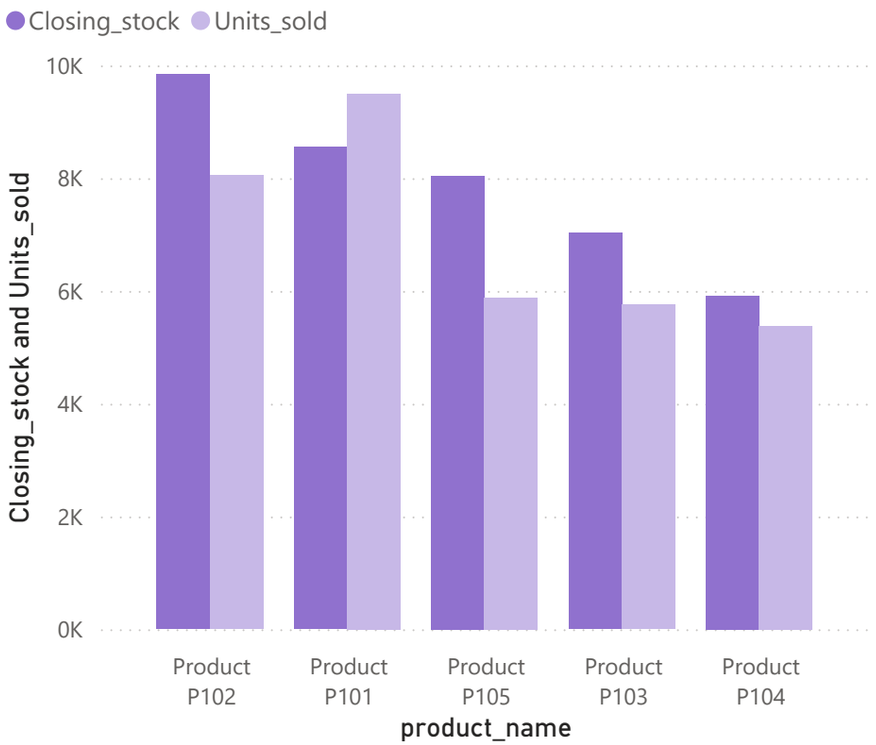
7	35K	39K	75.00%	66.99%	94.11	5.73
Stockout Count	Total Units Sold	Total Inventory	On-Time Delivery %	Forecast Accuracy %	Inventory Turnover Ratio	Average Lead Time

Forecasted vs Actual Demand



Closing_stock and Units_sold by product_name



supplier_name, product_category, product_name

- Supplier A
  - Clothing
  - Electronics
  - Home
  - Pharma
- Supplier B
- Supplier C
- Supplier D

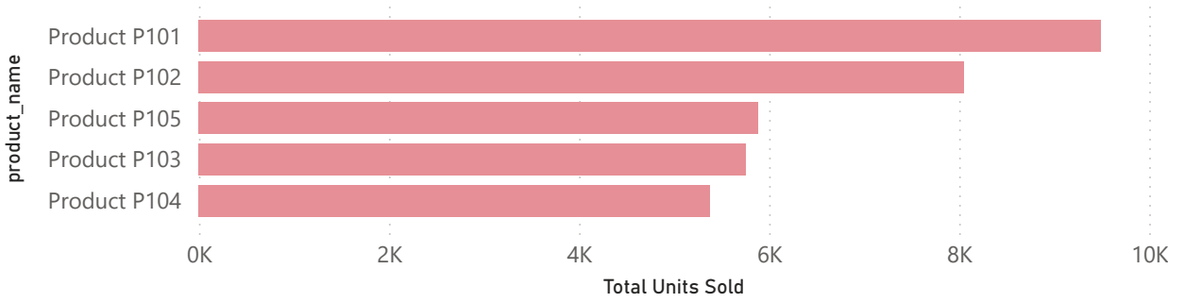
warehouse_location

- Bangalore
- Delhi
- Kolkata
- Mumbai

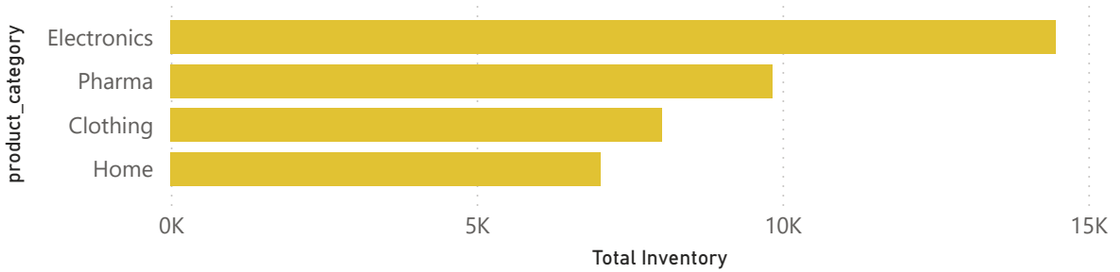
month

- 1
- 2
- 3
- 4

Total Units Sold by product_name

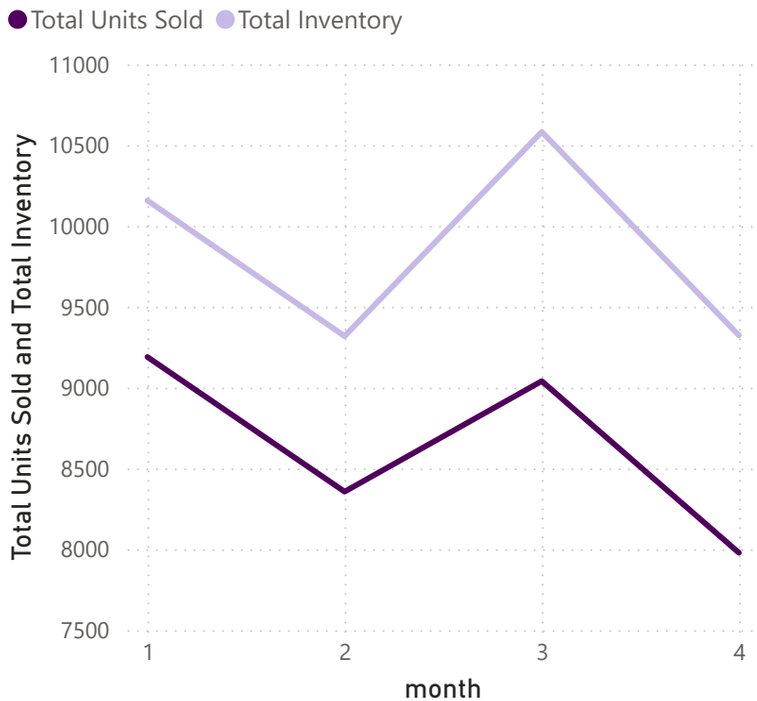


Inventory by Product Category



Total Units Sold by warehouse_location

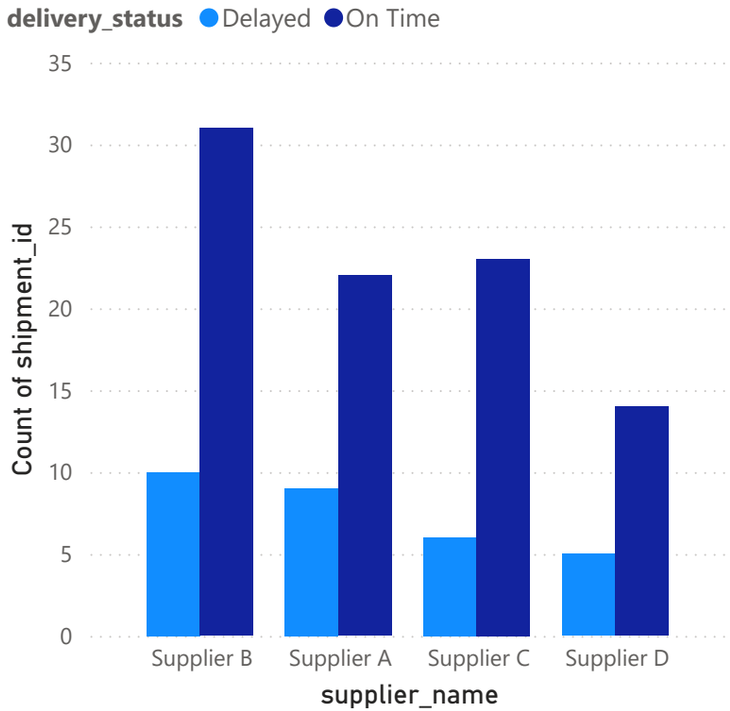
Monthly Inventory and Sales Trend



supplier_name, product_category, product_name

- Supplier A
  - Clothing
  - Electronics
  - Home
  - Pharma
- Supplier B
- Supplier C
- Supplier D

Supplier Delivery Performance



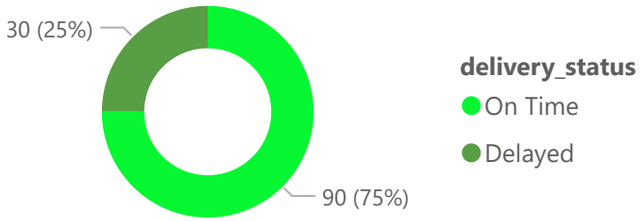
warehouse_location

- Bangalore
- Delhi
- Kolkata
- Mumbai

month

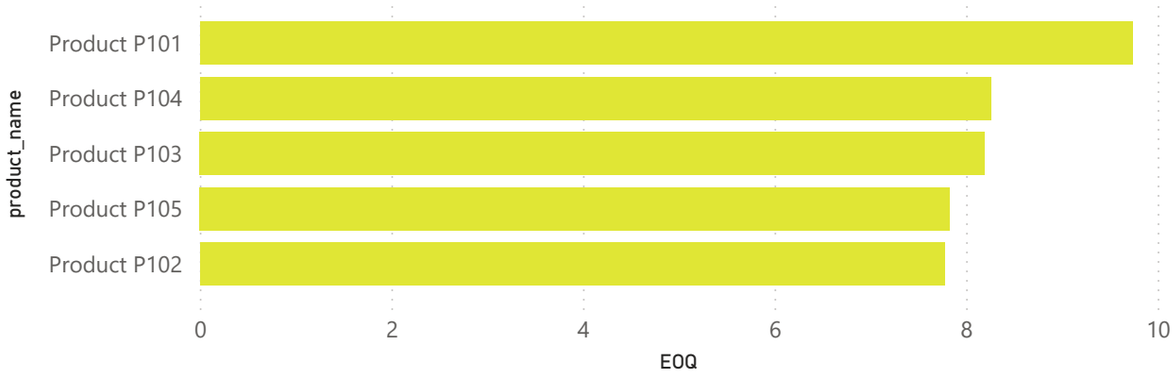
- 1
- 2
- 3
- 4

Delivery Status Distribution



product_name	Sum of closing_stock	Sum of opening_stock	Max of reorder_level	First delivery_status	Average of lead_time_days	First stockout_status
Product P101	8553	11901	292	Delayed	5.84	No
Product P102	9843	11755	294	Delayed	5.36	No
Product P103	7029	8284	294	Delayed	5.70	No
Product P104	5915	7426	293	Delayed	5.42	No
Product P105	8040	9394	294	Delayed	6.36	No

EOQ by product_name



product_name	Safety Stock	Reorder Point (Calculated)	Average of reorder_level	Stock Alert
Product P101	115.48	577.42	218.13	Reorder Needed
Product P102	89.84	449.22	222.54	Reorder Needed
Product P105	78.02	390.10	226.05	Reorder Needed
Product P103	68.33	341.64	228.20	Reorder Needed
Product P104	60.76	303.80	231.16	Reorder Needed

Key Business Insights & Recommendations

• **Understocking Risk Across Products:**
The calculated Reorder Point (ROP) is consistently higher than existing reorder levels across all products, indicating that the current inventory policy underestimates demand during lead time. This exposes the system to frequent stockouts.

• **Critical Products Requiring Attention:**
Products such as **P101 and P105** exhibit the highest stock gap between ROP and reorder levels, making them the most vulnerable to stock shortages and service disruptions.

• **Presence of Slow-Moving and Inefficient Inventory:**
Certain products show relatively high closing stock but lower sales, indicating slow-moving inventory. This ties up working capital and signals overstocking in specific segments.

• **Stockout Patterns Highlight Planning Gaps:**
Frequent stockouts in selected products suggest that reorder decisions are not aligned with actual demand variability and consumption patterns.

• **Demand Forecasting Inefficiency:**
The gap between forecasted demand and actual demand indicates inconsistencies in demand planning, which contributes to both excess inventory and stock shortages.

• **Supplier Performance Influences Inventory Levels:**
Variability in lead times and delayed deliveries from suppliers increases uncertainty, forcing higher safety stock and impacting inventory efficiency.

• **Inventory Turnover Imbalance:**
Differences in inventory turnover across products suggest uneven inventory utilization, with some products moving efficiently while others remain idle.

• **Safety Stock Buffer Not Fully Optimized:**
Although safety stock provides a buffer against uncertainty, it may not be adequately calibrated for high-demand or high-variability products.

• **Opportunity for Data-Driven Inventory Optimization:**
Aligning reorder levels with calculated ROP, improving forecasting accuracy, and enhancing supplier reliability can significantly reduce stockouts while optimizing inventory costs.

Overall Strategic Insight

The analysis highlights the need to transition from static inventory rules to a dynamic, data-driven supply chain strategy that balances service levels with cost efficiency.